

PRODUCT REQUIREMENTS DOCUMENT

PROJECT: TACTICAL PITCH SIMULATOR

1. Executive Summary

This project outlines the development of an MVP frontend validation prototype for a browser-based football management simulator. The objective is to build a fluid, interactive Single Page Application (SPA) that demonstrates complex state management and dynamic UI theming. The application must feel premium, utilizing fluid animations without relying on backend architecture.

2. Asset Provisioning & Data Integrity

The developer is provided with reference materials to dictate the absolute boundaries of the layout and the sole source of truth for the application data.

- **UI Wireframe (reference_image.png):** Mandates the spatial layout. A persistent sidebar must occupy exactly 250px on the left, with the main dynamic canvas set to a #121212 dark theme background.
- **Dataset Requirement:** The developer must manually create a `teams.jsx` file inside the `/src/data/` directory using the 20-team JSON array provided in **Appendix A** of this document. *No data structure alterations are permitted.*

3. Core Architectural Requirements

Framework / Library	Purpose & Constraint
React 18+ & Vite	Strictly vanilla JS (No TypeScript). No Server-Side Rendering (Next.js is forbidden).
React Router DOM	Must handle seamless client-side routing across 3 views without browser reloads.
Framer Motion	Mandatory for all page transitions (<AnimatePresence>) and interaction hover states.

4. Functional Scope & Routing

1. **Dashboard (/):** A landing view rendering a grid of all 20 clubs mapped from the dataset. Each club card must feature a Framer Motion `whileHover={{ scale: 1.05 }}` interaction.
2. **Locker Room (/squad):** An interactive squad builder rendering exactly 11 empty slots mapped into a 4-3-3 visual formation. The global state must strictly cap the active squad array at a maximum of 11 players. The UI must disable the action to simulate a match until exactly 11 players are slotted.
3. **Match Day (/simulation):** A static output screen that calculates and displays a win probability percentage based on the selected squad.

5. Dynamic Theming Engine

CRITICAL REQUIREMENT: No brand colors or logos may be hardcoded into the CSS files. They must be dynamically extracted.

The global state must track the user's `activeClub`. Upon selection, the application must immediately:

- Extract the `primary_hex` string from the selected club's object and apply it as the background color of the persistent 250px left Sidebar.
- Extract the `logo_path` string and render the image at exactly 64x64px in the top-left header component.

6. Out of Scope (Forbidden Elements)

- Backend server logic (Node/Express, Python) and database integrations (Firebase, Supabase, MongoDB).
- Authentication layers, complex multiplayer networking, or WebSockets.
- Heavy CSS gradients, drop shadows, or generic Bootstrap components.

7. Deliverable Specifications

The final handoff must be a Zipped Source Code archive (.zip). The `node_modules` folder must be explicitly excluded from the zip. The file must adhere strictly to the following naming convention:

`[PREFIX] - [item-name] - [version]_[YYYY-MM-DD].zip`

Appendix A: Required Dataset (teams.jsx)

The following array must be used exactly as written, with no structural modifications.

```
export const teams = [
  {
    name: "Arsenal",
    primary_hex: "#EF0107",
    secondary_hex: "#063672",
    logo_path: "https://placeholder.co/128x128/EF0107/FFFFFF?text=ARS"
  },
  {
    name: "Manchester City",
    primary_hex: "#6CABDD",
    secondary_hex: "#1C2C5B",
    logo_path: "https://placeholder.co/128x128/6CABDD/FFFFFF?text=MCI"
  },
  {
    name: "Liverpool",
    primary_hex: "#C8102E",
    secondary_hex: "#00B2A9",
    logo_path: "https://placeholder.co/128x128/C8102E/FFFFFF?text=LIV"
  },
  {
    name: "Manchester United",
    primary_hex: "#DA291C",
    secondary_hex: "#FBE122",
    logo_path: "https://placeholder.co/128x128/DA291C/FFFFFF?text=MUN"
  },
  {
    name: "Chelsea",
    primary_hex: "#034694",
    secondary_hex: "#EE242C",
    logo_path: "https://placeholder.co/128x128/034694/FFFFFF?text=CHE"
  },
  {
    name: "Tottenham Hotspur",
    primary_hex: "#132257",
    secondary_hex: "#FFFFFF",
    logo_path: "https://placeholder.co/128x128/132257/FFFFFF?text=TOT"
  },
  {
    name: "Newcastle United",
    primary_hex: "#241F20",
    secondary_hex: "#FFFFFF",
    logo_path: "https://placeholder.co/128x128/241F20/FFFFFF?text=NEW"
  },
  {
    name: "Aston Villa",
```

```

    primary_hex: "#670E36",
    secondary_hex: "#95BFE5",
    logo_path: "https://placeholder.co/128x128/670E36/FFFFFF?text=AVL"
  },
  {
    name: "Everton",
    primary_hex: "#003399",
    secondary_hex: "#FFFFFF",
    logo_path: "https://placeholder.co/128x128/003399/FFFFFF?text=EVE"
  },
  {
    name: "West Ham United",
    primary_hex: "#7A263A",
    secondary_hex: "#1BB1E7",
    logo_path: "https://placeholder.co/128x128/7A263A/FFFFFF?text=WHU"
  },
  {
    name: "Real Madrid",
    primary_hex: "#FFFFFF",
    secondary_hex: "#FEBE10",
    logo_path: "https://placeholder.co/128x128/FFFFFF/000000?text=RMA"
  },
  {
    name: "FC Barcelona",
    primary_hex: "#004D98",
    secondary_hex: "#A50044",
    logo_path: "https://placeholder.co/128x128/004D98/FFFFFF?text=FCB"
  },
  {
    name: "Atletico Madrid",
    primary_hex: "#CB3524",
    secondary_hex: "#272E61",
    logo_path: "https://placeholder.co/128x128/CB3524/FFFFFF?text=ATM"
  },
  {
    name: "Bayern Munich",
    primary_hex: "#DC052D",
    secondary_hex: "#0066B2",
    logo_path: "https://placeholder.co/128x128/DC052D/FFFFFF?text=FCB"
  },
  {
    name: "Borussia Dortmund",
    primary_hex: "#FDE100",
    secondary_hex: "#000000",
    logo_path: "https://placeholder.co/128x128/FDE100/000000?text=BVB"
  },
  {
    name: "Paris Saint-Germain",
    primary_hex: "#004170",
    secondary_hex: "#DA291C",
    logo_path: "https://placeholder.co/128x128/004170/FFFFFF?text=PSG"
  },
  {

```

```
    name: "Juventus",
    primary_hex: "#000000",
    secondary_hex: "#FFFFFF",
    logo_path: "https://placeholder.co/128x128/000000/FFFFFF?text=JUV"
  },
  {
    name: "AC Milan",
    primary_hex: "#FB090B",
    secondary_hex: "#000000",
    logo_path: "https://placeholder.co/128x128/FB090B/FFFFFF?text=ACM"
  },
  {
    name: "Inter Milan",
    primary_hex: "#010E80",
    secondary_hex: "#000000",
    logo_path: "https://placeholder.co/128x128/010E80/FFFFFF?text=INT"
  },
  {
    name: "Napoli",
    primary_hex: "#12A0D7",
    secondary_hex: "#FFFFFF",
    logo_path: "https://placeholder.co/128x128/12A0D7/FFFFFF?text=NAP"
  }
];
```